Ultrasound-based Human–Machine Interfacing for Soft Hand Exoskeletons under Assisted and Passive Conditions

Bruno Grandi Sgambato[1†], Juan C. Maldonado-Mejía[2†], Tristan Wen Jie Choo[1], Marcela Múnera[4], *Member, IEEE*, Camilo A. R. Díaz[2], *Member, IEEE*, Majid Taghavi[3], Dario Farina[1], *Fellow, IEEE*, and Carlos A. Cifuentes[4], *Senior Member, IEEE*

Abstract*— Soft hand exoskeletons are promising devices for rehabilitation and assistance in daily activities. However, their effectiveness depends on robust human–machine interfaces (HMIs) capable of reliably decoding user intent across a variety of conditions. Amplitude mode (A-mode) ultrasound (US) has been demonstrated as a strong candidate for HMI control. Yet, it remains unclear how exoskeleton activation, which mechanically deforms the same muscles being monitored, affects the US signals used for intention decoding. In this study an adapted commercial soft pneumatic hand exoskeleton was used and the sensitivity of A-mode ultrasound signals was analysed in two phases and a total of four scenarios: (i) *Control phase*, user's voluntary movement without exoskeleton; and the *Main phase* with (ii) *Passive*, exoskeleton actuated without voluntary motion; (iii) *Active*, voluntary motion without exoskeleton actuation; and (iv) *Combined*, voluntary motion assisted by exoskeleton actuation. Five healthy participants were included in the study, performing seven grasps inspired by a standardized assessment protocol. The mean cross-validation accuracy in classifying grasp type from US signals was on average 0.83 for *Control*, 0.72 for *Active* and *Combined*, and only 0.34 for *Passive*. These results, together with feature quantification analysis, suggested that voluntary muscle activation dominates the US signal during exoskeleton use, while passive motions determined by the glove produced weak and inconsistent US changes. Consequently, A-mode US appears to be suitable for detecting users' intentions, even in the presence of soft exoskeleton assistance.

I. INTRODUCTION

Motor impairment of the hands is a common consequence of neurological injuries, such as stroke [1]. In these conditions, individuals experience difficulty in performing activities of daily living (ADLs), which affects their independence and quality of life. Accordingly, intense and continuous therapeutic exercises are important for patients to regain hand motor function and achieve successful rehabilitation. In this context, robotic devices like soft hand exoskeletons have gained increasing attention [2]. Within soft hand exoskeletons, tendon-driven systems and pneumatic architectures represent two of the most prominent and widely explored actuation paradigms [3], both having widely demonstrated their efficacy in supporting rehabilitation tasks and assisting users during ADLs [4], [5].

To achieve appropriate control and assistance, the design of robust human-machine interfaces (HMIs) is a priority during exoskeleton development [6]. These interfaces focus on translating the user's desired movements or commands into actions for the device, while also providing feedback and natural control. Surface electromyography (sEMG) is one of the most widely used modalities to decode the human body's movement intention and muscle activity, and it has been successfully applied to control upper-limb prostheses and exoskeletons [7], [8]. However, sEMG signals are sensitive to variations in electrode-skin impedance due to changes in temperature and sweating, arm posture, and crosstalk between adjacent muscles [6], [8]. These limitations compromise its control performance and robustness, making long-term use challenging, especially in real-world conditions.

Ultrasound (US) has recently emerged as an HMI modality for controlling upper-limb devices, offering an alternative to sEMG [9]. Instead of measuring electrical activity, US probes image tissue morphology by transmitting acoustic waves and receiving the echoes from tissue boundaries. Variations in echo amplitude and timing therefore reflect local deformation caused by both in superficial and deep muscles, not accessible with surface electrodes. Among US acquisition methods, Brightness mode (B-mode) and Amplitude mode (A-mode) are the most commonly used as HMIs. In B-mode, an array of transducers is used to reconstruct 2D cross-sectional images of muscles, and has been successfully used to classify hand gestures, predict finger position, and control virtual hands or prostheses in real-time [10], [11]. However, B-mode systems are typically bulky, heavy, and the rigid probe geometry complicates integration into lightweight devices such as an exoskeleton.

A-mode US, on the other hand, uses single-element transducers to detect echoes along a one-dimensional lines, trading off fine lateral resolution for improved wearability, larger field of view, and lower processing requirements [12]. A-mode systems have recently shown feasibility as HMIs for prosthesis [12] and shown very strong cross-participant performance for virtual reality control [13]. In these applications, however, there is no direct coupling between the HMI outputs and the US signals being recorded. While

† These authors contributed equally to this work

This work was supported by The VIVO Hub for Enhanced Independent Living, UKRI and EPSRC, under Grant UKRI142, and was financed in part by CAPES (001), FAPES, and CNPq (306323/2024-9 and 402739/2024-8).

[1] B.G.S, T.W.J.C, and D.F are with the Department of Bioengineering, Imperial College, U.K.

[2] J.C.M.M and C.A.R.D are with the Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil.

[3] M.T. is with the School of Engineering and Materials Science, Queen Mary University of London, U.K.

[4] M.M and C.A.C. are with the Bristol Robotics Laboratory, University of the West of England, Bristol, U.K. carlos.cifuentes@uwe.ac.uk

for all modalities (e.g., sEMG) assistance causes users to adapt and change their motor strategies, US signals are also altered as a direct mechanical consequence of the mechanical assistance at the joints. Though works have demonstrated US capabilities on controlling exoskeletons [14], [15], the effects of this coupling have not been explicitly investigated.

Therefore, this work investigates whether tissue deformation driven by a soft pneumatic hand exoskeleton significantly alters A-mode US signals and if this has the potential to affect device control. To this end, A-mode US data were collected during functional grasps under two phases and a total four scenarios: (i) *Control phase*, user's voluntary movement without the exoskeleton; *Main phase* with (ii) *Passive*, exoskeleton actuating and no user's voluntary effort; (iii) *Active*, user's voluntary movement with an unactuated exoskeleton; and (iv) *Combined*, user's voluntary movement assisted by the actuated exoskeleton. Using this dataset, the behaviour of simple gesture classifiers under cross-validation with and without assistance was explored. In addition, variations in the features extracted in each scenario were quantified using a Mahalanobis-distance analysis in the feature space.

II. METHODOLOGY

A. Participants and Ethics

Five healthy participants (5 male, 28.6 ± 1.5 years old) were recruited for this study. They were briefed on the experimental procedures, presented with a participant information sheet, allowed to ask any questions and asked to sign a consent form. Only then experiments started. The acquired datasets were not registered in a public database but can be made available, as appropriate, upon request. All procedures and experiments were approved in accordance with the declaration of Helsinki by the Imperial College Research Ethics Committee (ref: 22IC7602).

B. Soft Hand Exoskeleton

The hand exoskeleton is an adaptation of a commercial exoskeleton (Syrebo, China). The device includes a pneumatic system with two air pumps and one solenoid electrovalve. The system was adapted with a network of five solenoid electrovalves (Adafruit Industries LLC, USA), to control each finger separately. Thus, flexion and extension movements are possible, and therefore, the performance of different grasp types. For control of the pneumatic system, switches based on MOSFET transistors (SI9956DY, Vishay Siliconix, USA) and optocouplers (MCT62S, Onsemi, USA) were used. For processing and control of the device, a single-board computer (Raspberry Pi 4 Model B) was used. All components were housed in a custom 3D-printed control box.

C. Ultrasound Hardware and Acquisition

A portable US acquisition system (MoUSE, Fraunhofer IBMT, Sulzbach, DE) was used to drive US acquisition [16]. The system was connected to 12 4-MHz piezoelectric single element transducers (circular, 11 mm radius). US echoes were sampled at 50 MSPS with 12-bit resolution for 110 μs (5500 samples per line) at approximately 20 frames per second. The single-transmit multi-receive approach used in [12] was also implemented in this work. Briefly, it follows similarly from the Full Matrix Capture procedure for phased-arrays, where all combinations of single element transmit-receive are sampled per frame.

The 12 A-mode transducers were arranged in two identical bracelets. Each bracelet was composed of six 3D printed enclosures linked together by a flat elastic string. Bracelets were placed on each participant's forearm with sensors in each bracelet adjusted to be roughly equidistant from their neighbours. On all participants, the two bracelets were positioned in the middle of the forearm, at 5 cm and 10 cm distally from the antecubital fossa, respectively.

D. Experimental Procedures

Participants joined a single recording session that lasted approximately 45 minutes. They were asked to sit in a chair facing a table and a computer screen, and to position their left arm on a soft cushion atop the table and maintain a comfortable but stable position. On the screen, participants were shown a custom interface that guided them to perform hand gestures based on cues. The complete experimental setup is shown in Fig. 1. The interface prompted users to perform each gesture in sequence. A yellow border around an image of a gesture indicated a preparation time in which users (or the powered glove, depending on the situation) should move from the current position to the target gesture. Thereafter, a green border indicated that ultrasound data was being recorded, and therefore that users should hold the target gesture in a stable manner. The yellow border period lasted between two to six seconds depending on the situation while the green border period always lasted for four seconds.

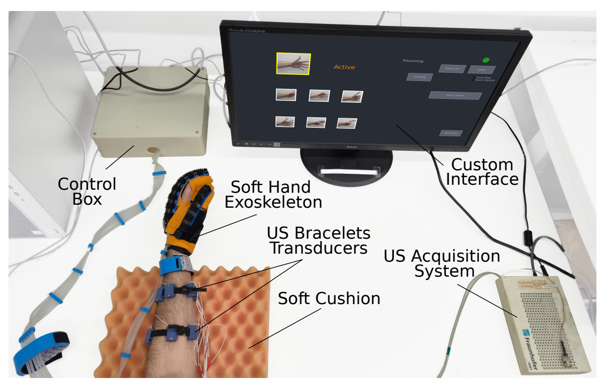

Fig. 1. Experimental setup for the test. Participants were seated facing a table and monitor, with the left arm supported on a soft cushion to maintain a comfortable and stable posture during the experiment.

Seven gestures (Rest, Power Grasp, Pinch, Hook Grasp, All Finger Extension, Thumb Only Grip, Index Point) from the AHAP (Anthropomorphic Hand Assessment Protocol) [17] were selected for this work. These were chosen based on the common combinations of finger flexion/extension used during ADLs. For the purposes of this work, only motions of the fingers were considered (i.e. the Hook gesture did not include a wrist ulnar deviation).

The experiment consisted in two phases. A short *Control phase*, performed while the participant was not wearing the exoskeleton, was followed by the *Main phase* with the exoskeleton being worn. The *Control phase* consisted of a single recording period with five repetitions of each gesture, performed in sequence, one immediately after the other. After this first phase, the exoskeleton was positioned on each participant's left hand and turned on.

The *Main phase* consisted of five recording periods with one-minute breaks in between. Recording periods during this phase differed depending on the state of the powered glove. During each of the five recording periods of the *Main phase*, *Passive (Pa)*, *Active (Ac)*, and *Combined (Co)* repetitions were performed without breaks. The order of these three types of repetitions was randomized. The graphical interface the mode used in each set of repetitions. On *Passive* repetitions, participants were instructed not to perform any voluntary movements, keep their hands relaxed, and allow the exoskeleton to perform the gestures by itself. *Active* repetitions were the opposite, and participants were informed that the glove would not be powered and that they should volitionally contract their hands (against the elastic resistance of the glove) to perform the gestures being shown. Lastly, for the *Combined* repetitions, participants were informed that the glove would be actuated to carry out the gestures, but that they should still voluntarily engage their own muscles to perform the gestures themselves. The four experimental conditions and the seven gestures are summarised in Fig. 2.

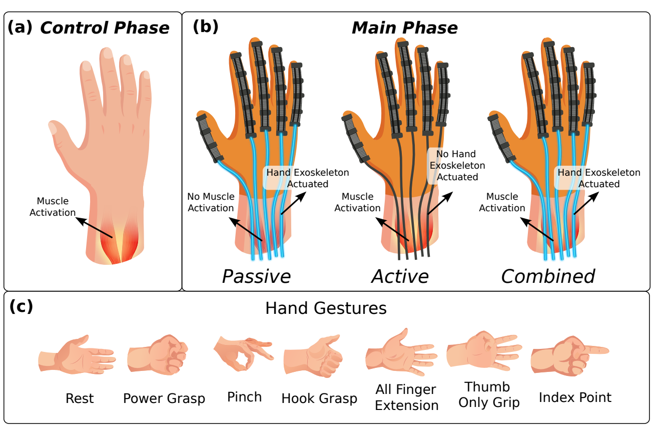

Fig. 2. Experimental conditions and gestures. (a) *Control phase* (user's voluntary movement without exoskeleton). (b) *Main phase* with: *Passive* (exoskeleton actuation without user's voluntary movement), *Active* (user's voluntary movement without exoskeleton actuation), and *Combined* (user's voluntary motion + exoskeleton actuation). (c) The seven functional grasps used and inspired by the AHAP.

Before the start of this phase, participants experienced how the glove contracted their fingers to perform the gestures. They were then explicitly instructed to try to replicate the way the exoskeleton performed each gesture in *Passive* sessions when they were volitionally performing the gestures in the *Active* session.

Overall, each participant performed five repetitions of each of the seven gestures, in each of the four conditions, for a total of 140 gestures per participant. With each recording lasting approximately 4 seconds, this resulted in slightly more than 9 minutes, or approximately 11 thousand frames of US data per participant.

E. Data Processing and Analysis

The acquired US data was processed following HMI interfacing pipelines developed and used in previous works [12], [18]. Radiofrequency (RF) data from each single element transducer/channel was time gain compensated at eight equal-sized depth stages following $G = 3 + 3 * n$ with n ranging for 1 to 8. Subsequently, the RF data was band-pass filtered between 3.6 and 4.4 MHz and the first 800 (out of 5500) samples were removed due to transducer and static noise artifacts. The data was then downsampled by a factor of five (to 10 MSPS) and Hilbert enveloped. The Root Mean Square feature was then extracted from each receive channel over non-overlapping sliding windows (40 samples wide). Due to the high dimensionality of the feature space extracted with this method (1656 features per frame), feature dimensionality was further reduced by Principal Component Analysis (PCA) to the first 100 components (explaining approximately 0.99 ± 0.01 of the variance, on average).

The PCA-extracted components together with the synchronized gesture cues were then used for the analysis. The data from each participant was analysed independently. Two sets of analyses were conducted; these were aimed at exploring the similarities and differences between passive and active movements, as well as understanding the impact of these different movements in machine learning-driven intention detection and gesture classification.

The first analysis involved evaluating the performance of a simple multiclass Support Vector Machine (SVM) Classifier (one-versus-one, radial basis function kernel with $C = 1$ [19]) under different cross-validation scenarios. *Control* performance results were obtained from the glove-off recordings obtained in the *Control phase* with a five-fold leave-one-out cross-validation strategy. Baseline, glove-on, results from the *Main phase* were also obtained using five-fold leave-one-out cross-validation performed independently for each recording situation (*Passive*, *Active* and *Combined*). Additionally, all combinations of cross-validations across the different types of movements were also tested to give a total of nine cross-validation situations. Here, similar to the manner in which the baseline results were calculated, different combinations of four (out of five) repetitions of the contraction types were used for training, while each of the five repetitions from the contraction type to be validated were used for testing. As the dataset is balanced with approximately the same amount of frames for each gesture, accuracy was chosen to be the main performance metric.

The second analysis involved evaluating the feature space from the acquired US data. Based on the methods proposed by [20], [21], two metrics derived from the Mahalanobis distance were calculated to quantify the repeatability and separability of the distributions, and one metric derived from the covariances to quantify the magnitude of the variances each recording situation.

The Repeatability Index (RI) was used to quantify the intra-class distances between a training and testing distribution. Bunderson *et al.* [20] defined this index as one-half of the average, across gestures, and Mahalanobis distance between the training and testing groups. Thus, the RI was calculated using the Eq. 1, where μ_{tri} and μ_{tei} are the US feature vector centroids, of class i, for the training and testing folds respectively; Σ_{tri}^{-1} is the covariance matrix, for class i, of the training fold; and C represents the total number of classes. The RI therefore expresses how distant, on average, the features from each gesture from the training fold are in relation to the testing fold while taking in consideration the distribution (in the form of the covariance matrix of the training fold). A larger RI implies a larger separation between folds, and therefore a lower repeatability in that scenario. The RI however, only takes into account intra-gesture distances, without considering the inter-gesture separability.

$$RI = \frac{1}{C}\sum_{i=1}^{C}\left(\frac{1}{2}\sqrt{\left(\mu_{tri}-\mu_{tei}\right)^{T}\Sigma_{tri}^{-1}\left(\mu_{tri}-\mu_{tei}\right)}\right), \quad (1)$$

The Separability Index (SI) [20], on the other hand, accounts for the similarities across gestures. The SI is one-half of the average, across gestures, minimum Mahalanobis distance between each training gesture and all other testing gestures. The Eq. 2 was used to determine the SI, where μ_{trj} and μ_{tei} are the US feature vector centroids, of classes j and i, for the training and testing folds respectively; Σ_{trj}^{-1} is the covariance matrix, for class j, of the training fold; and C represents the total number of classes. The SI therefore expresses the average proximity between each gesture and its nearest neighbouring gesture. Lower SI represent lower separability between classes. This work report a ratio between SI and RI (SI/RI) as a combined measure of both the between-class (RI) and within-class (SI) clusterization, similarly to the results of Jia *et al.* [21]. A high SI/RI therefore denotes that a certain more separated and compact class clusters.

$$SI = \frac{1}{C}\sum_{i=1}^{C}\left(\min_{j=1,j\neq i}^{C}\frac{1}{2}\sqrt{\left(\mu_{trj}-\mu_{tei}\right)^{T}\Sigma_{trj}^{-1}\left(\mu_{trj}-\mu_{tei}\right)}\right), \quad (2)$$

Lastly, the Mean Semi-Principal Axis (MSA) [20] was calculated and describes the size of the hyperellipsoid that encompasses the training distributions. It was defined as the average, across gestures, of the geometric mean of the semi-principal axes (a_k) of the features of the training distribution. The Eq. 3 describes how to obtain the MSA, where Λ is diagonal matrix of eigenvalues ($\lambda_1, ..., \lambda_D$) from the eigendecomposition of the covariance matrix of the training data Σ_{tri} for class i; n represents the number of standard deviations encompassed by the hyperellipsoid (defined as 2 for this work); and C represents the total number of classes while D represents the total number of features. The MSA therefore represents the variability of the training data, with lower values suggesting gestures clusters are less spatially separated from each other.

$$MSA = \frac{1}{C}\sum_{i=1}^{C}\left(\left(\prod_{j=1}^{D}a_{ik}\right)^{1/D}\right), \quad (3)$$
$$a_{ik} = \sqrt{n * \Lambda}$$

RI, SI, and MSA were calculated via cross-validation on the same folds used for the accuracy calculation to ensure consistency. SI and MSA are only meaningful for the three baseline situations and were therefore only calculated for it.

F. Statistics

Results for the SI and MSA were aggregated across all participants, and the distributions obtained for the three tested cases (*Pa-Pa*, *Ac-Ac*, *Co-Co*) were subjected to a one-way ANOVA evaluation. If significance ($p < 0.05$) was found, a post-hoc Tukey–Kramer was used to compare the means of each distribution to each other. Across the text, results are reported as mean $\pm$ standard deviation.

III. RESULTS

A. Cross-Validation Performance

Table I includes accuracy of the cross-validation results for all participants, across all tested situations. It is highlighted that the chance level, for a 8-class classification problem, is approximately 0.13. Across participants the *Control* accuracy was 0.83 ± 0.14, while results of 0.34 ± 0.16, 0.72 ± 0.14, and 0.72 ± 0.17 were found for the *Passive*, *Active* and *Combined* baseline cases, respectively. Average results for the other cross-validation situations were varied but overall all situations involving testing on, or only training on, the *Passive* dataset showed low results, close to chance (maximum of 0.29 ± 0.09 for training of *Passive*, testing on *Combined*). While results for the other cross-validations groups were all higher than 0.64. Fig. 3 shows results for the *Pa-Pa*, *Ac-Ac*, *Ac-Pa*, and *Ac-Co* cross-validations to exemplify performance results on a per gesture basis.

B. Ultrasound Feature Quantification Analysis

Fig. 4a reports the RI for the three baseline situations (*Pa-Pa*, *Ac-Ac*, *Co-Co*) and for the three possible (non-repeating) combinations (*Pa-Ac*, *Pa-Co*, *Ac-Co*). Results for situations including the *Passive* repetitions (*Pa-Pa*, *Pa-Ac*, and *Pa-Co*) showed notably higher RI when compared with the situations that did not include *Passive* repetitions, showcasing lower repeatability between the *Passive* and non-passive scenarios. On the other hand, the RIs for the *Ac-Co* situation were similar to the non-passive baselines (*Ac-Ac*, *Co-Co*).

Fig. 4b includes the SI, aggregated for all participants, for the three baseline situations. The distributions for each situation were not statistically different from each other. Conversely, Fig. 4c shows that the MSA for *Passive* repetitions was significantly lower than the MSA for the *Active* and *Combined* repetitions, suggesting that while clusters centroids are equally distant in all cases, the features on the *Passive* cluster are significantly less variant.

TABLE I

Cross-Validation Accuracy showing mean $\pm$ standard deviation for Pa: Passive, Ac: Active, Co: Combined

	Control	Baseline			Passive		Active		Combined		Pa + Ac	Pa + Co	Ac + Co
		Pa	Ac	Co	Ac	Co	Pa	Co	Pa	Ac	Co	Ac	Pa
P1	0.85 ± 0.14	0.35 ± 0.06	0.77 ± 0.14	0.65 ± 0.27	0.15 ± 0.09	0.27 ± 0.11	0.12 ± 0.06	0.61 ± 0.11	0.26 ± 0.1	0.65 ± 0.19	0.67 ± 0.16	0.66 ± 0.1	0.2 ± 0.11
P2	0.85 ± 0.11	0.41 ± 0.2	0.72 ± 0.11	0.82 ± 0.15	0.41 ± 0.06	0.41 ± 0.01	0.33 ± 0.15	0.71 ± 0.07	0.19 ± 0.05	0.75 ± 0.15	0.72 ± 0.06	0.76 ± 0.12	0.24 ± 0.05
P3	0.93 ± 0.08	0.36 ± 0.2	0.67 ± 0.27	0.69 ± 0.08	0.31 ± 0.06	0.42 ± 0.18	0.22 ± 0.12	0.5 ± 0.12	0.34 ± 0.23	0.62 ± 0.23	0.62 ± 0.13	0.54 ± 0.16	0.33 ± 0.21
P4	0.73 ± 0.2	0.32 ± 0.13	0.67 ± 0.12	0.74 ± 0.16	0.2 ± 0.11	0.19 ± 0.02	0.17 ± 0.04	0.64 ± 0.21	0.15 ± 0.08	0.7 ± 0.14	0.63 ± 0.19	0.6 ± 0.13	0.12 ± 0.08
P5	0.78 ± 0.15	0.29 ± 0.16	0.79 ± 0.07	0.69 ± 0.17	0.12 ± 0.12	0.18 ± 0.14	0.13 ± 0.02	0.72 ± 0.19	0.14 ± 0.08	0.89 ± 0.08	0.73 ± 0.15	0.83 ± 0.06	0.17 ± 0.06
Avg	0.83 ± 0.14	0.34 ± 0.16	0.72 ± 0.14	0.72 ± 0.17	0.24 ± 0.09	0.29 ± 0.09	0.19 ± 0.08	0.64 ± 0.14	0.22 ± 0.11	0.72 ± 0.16	0.67 ± 0.14	0.68 ± 0.11	0.21 ± 0.10

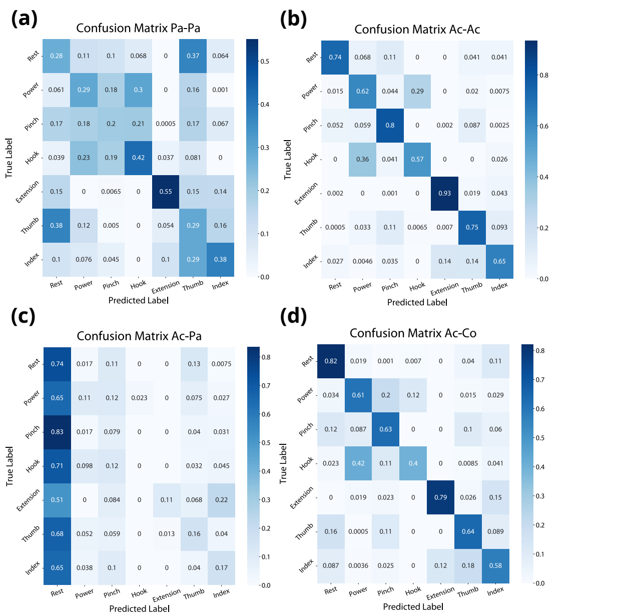

Fig. 3. Confusion Matrices of the accuracies for the Passive-Passive **(a)**, Active-Active **(b)**, Active-Passive **(c)**, and Active-Combined **(d)** scenarios. Results are expressed in percentages aggregated across the 5-fold leave-one-out cross validation, approx. 14,000 frames per matrix.

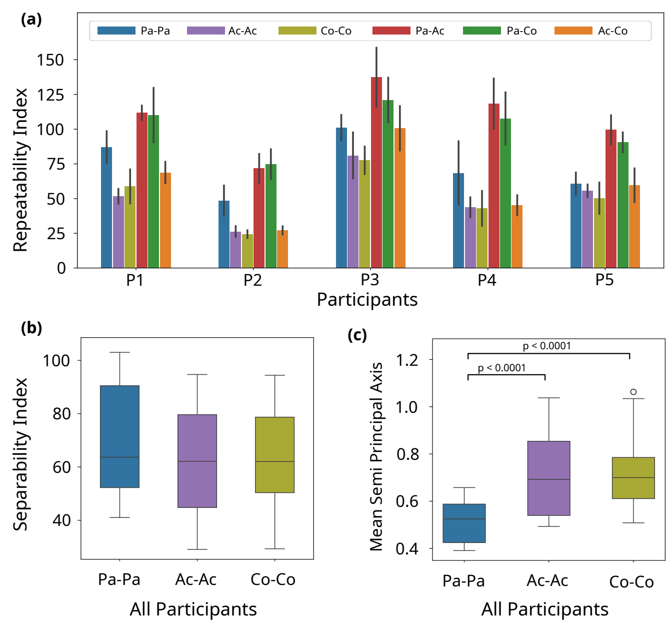

Fig. 4. Feature Quantification Results. **(a)** Repeatability Index of features per participant, across six cross-validation situations. Each bar shows the mean of the 5-fold xxxxx-validation ($n = 5$) with the error bar representing ± 1 standard deviation. Separability Index **(b)** and MSA **(c)** of features of all participants, across three cross-validation situations.

IV. Discussion

The achieved average control results (83%) are similar to values found in the literature (96% [22], 84% [23]). The closest comparison found in the literature for our Baseline *Active* results is the study by Zou *et al.* [15] that showed performance between $0.7 - 0.9$ R^2 when only using US. The relatively high cross-validation accuracies in Table I (> 0.69), where *Active* (Fig. 3b) and *Combined* movements are used in training and/or testing, suggest that US as a HMI modality is well suited to predicting the different hand gestures when voluntary control is involved.

Conversely, the lower cross-validation accuracies that are obtained when *Passive* movements are used imply that the muscle deformations produced solely by the glove are less consistent and harder to predict, as expected. The higher RI, and therefore lower repeatability, for the cross-validations involving *Passive* movements (*Pa-Ac*, *Pa-Co*), suggest that feature signatures from the *Passive* are notably distinct than the ones from non-passive movements. The lower RI for the *Ac-Co* situation, conversely, indicates that when both the *Passive* and *Active* components are present, the gesture signature is mainly determined by the *Active* component (seen on Fig. 3d). This may be explained by the fact that the different movements the glove produces might not be large enough to result in significantly different deformations of the forearm muscles that are easy to distinguish from one another. While volitional control starts by motor unit activation and muscular deformation the glove produces deformations of the forearm muscles in an indirect, top-down manner, by acting on the tendons, which then cause the muscles to deform. Therefore, these might only cause the associated tendons to stretch or become more slack rather than to deform the muscles. Predictions of mostly the rest gesture (Fig. 3c) for the *Ac-Pa* situation are evidence of this difference.

The results of the SI and MSA also provisionally support this conclusion. The absence of differences between the SI in each situation implies that, regardless of the origin of the contraction, the gestures are equally distant from each other, while the significantly lower MSA for the passive situation supports the idea that the magnitude of changes in characteristics is smaller when the gestural glove is activated. These results support the hypothesis that, unlike sEMG, US signals are directly affected by the mechanical actuation caused by exoskeletons, placing them closer to other sources of kinematic signals. However, there also appeared to be notable differences in the characteristics recorded depending on the source of actuation. The same gesture (same final kinematic position) changes the US characteristics differently depending on the source of actuation, and some *Passive* deformations generate little or no change in the signals.

Among the limitations of this work is that the experimental protocol only investigated static hand gestures without wrist or forearm movements. For practical real-world applicability, the wearability and real-time performance of the system must be thoroughly addressed. Thus, future work should focus other joints, where the interplay between passive and voluntary deformations could differ. With aid of extra kinematic measurements a regression task where the relative contributions of both the individual and the exoskeleton to the overall movement could also be explored. Lastly, this study explicitly removed the user adaptation factor from the equation by having users follow specific instruction and not adapt to assistance, real implementation would first require an investigation how user adaptation affects measured signals and control performance.

V. CONCLUSIONS

This study explored differences in the deformation of the forearm muscles as measured by A-mode US signals when produced solely by a soft hand exoskeleton (*Passive* movements), when performed volitionally (*Active movements*), and when elicited by a combination of the glove and voluntary movement (*Combined* movements). When evaluated across seven hand gestures in the five participants, the signal generated during *Combined* movements proved to be more similar to *Active* movements, due to higher cross-validation accuracies and lower RIs. *Passive* movements caused less variation on the US derived features, shown by lower MSA, with some gestures cross-validations having very low accuracies, close to random chance.

REFERENCES

[1] W. H. Organization, *World health statistics 2025: monitoring health for the SDGs, Sustainable Development Goals*. World Health Organization, 2025.

[2] Z. T. H. Tse, Y. Chen, S. Hovet, H. Ren, K. Cleary, S. Xu, B. Wood, and R. Monfaredi, "Soft robotics in medical applications," *Journal of Medical Robotics Research*, vol. 03, no. 03n04, p. 1841006, 2018.

[3] T. Shahid, D. Gouwanda, S. G. Nurzaman, and A. A. Gopalai, "Moving toward soft robotics: A decade review of the design of hand exoskeletons," *Biomimetics*, vol. 3, no. 3, 2018. [Online]. Available: https://www.mdpi.com/2313-7673/3/3/17

[4] H. Su, F. Masiero, F. Missiroli, M. M. El Sidani, C. Piazza, and L. Masia, "Soft wearable robotic kit for forearm rotation and grasping motion tracking based on embedded end-effector-level sensor system," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 18 256–18 261.

[5] P. Polygerinos, Z. Wang, K. C. Galloway, R. J. Wood, and C. J. Walsh, "Soft robotic glove for combined assistance and at-home rehabilitation," *Robotics and Autonomous Systems*, vol. 73, pp. 135–143, 2015, wearable Robotics.

[6] L. Guo, Z. Lu, and L. Yao, "Human-machine interaction sensing technology based on hand gesture recognition: A review," *IEEE Transactions on Human-Machine Systems*, 2021.

[7] L. Zongxing, Z. Jie, Y. Ligang, C. Jinshui, and L. Hongbin, "The human-machine interaction methods and strategies for upper and lower extremity rehabilitation robots: A review," *IEEE Sensors Journal*, vol. 24, no. 9, pp. 13 773–13 787, 2024.

[8] H. Zhou and G. Alici, "Non-invasive human-machine interface (hmi) systems with hybrid on-body sensors for controlling upper-limb prosthesis: A review," *IEEE Sensors Journal*, 2022.

[9] X. Yang, C. Castellini, D. Farina, and H. Liu, "Ultrasound as a neurorobotic interface: A review," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 54, no. 6, pp. 3534–3546, 2024.

[10] S. M. Engdahl, S. A. Acuña, E. L. King, A. Bashatah, and S. Sikdar, "First demonstration of functional task performance using a sonomyographic prosthesis: A case study," *Frontiers in Bioengineering and Biotechnology*, vol. 10, 2022.

[11] N. Akhlaghi, C. A. Baker, M. Lahlou, H. Zafar, K. G. Murthy, H. S. Rangwala, J. Kosecka, W. M. Joiner, J. J. Pancrazio, and S. Sikdar, "Real-time classification of hand motions using ultrasound imaging of forearm muscles," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 8, pp. 1687–1698, 2016.

[12] B. G. Sgambato, M. H. Hasbani, D. Y. Barsakcioglu, J. Ibáñez, A. Jakob, M. Fournelle, M.-X. Tang, and D. Farina, "High performance wearable ultrasound as a human-machine interface for wrist and hand kinematic tracking," *IEEE Transactions on Biomedical Engineering*, vol. 71, no. 2, pp. 484–493, 2024.

[13] B. Grandi Sgambato, B. K. Hodossy, D. Y. Barsakcioglu, X. Yang, A. Jakob, M. Fournelle, M.-X. Tang, and D. Farina, "Virtual reality interactions via a user-generic ultrasound human-machine interface for wrist and hand tracking," *Nat. Commun.*, 2025.

[14] L. Xing, Z. Hu, and Y. Yin, "Ultrasound-assisted intent recognition for precise control of lower limb exoskeletons," in *Proceedings of the International Conference on Biomedicine and Intelligent Technology*. Association for Computing Machinery, 2024.

[15] Y. Zou, L. Cheng, and Z. Li, "A multimodal fusion model for estimating human hand force: Comparing surface electromyography and ultrasound signals," *IEEE Robotics and Automation Magazine*, vol. 29, no. 4, pp. 10–24, 2022.

[16] M. Fournelle, T. Grün, D. Speicher, S. Weber, M. Yilmaz, D. Schöb, A. Miernik, G. Reis, S. Tretbar, and H. Hewener, "Portable ultrasound research system for use in automated bladder monitoring with machine-learning-based segmentation," *Sensors*, vol. 21, 2021.

[17] I. Llop-Harillo, A. Pérez-González, J. Starke, and T. Asfour, "The anthropomorphic hand assessment protocol (ahap)," *Robotics and Autonomous Systems*, vol. 121, p. 103259, 2019.

[18] B. G. Sgambato, H. Hakami, X. Yang, D. Y. Barsakcioglu, A. Jakob, M. Fournelle, A. H. McGregor, M.-X. Tang, and D. Farina, "Towards natural multi-dof prosthetic control with distributed ultrasound," in *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*, 2024, pp. 1–6.

[19] X. Yang, X. Sun, D. Zhou, Y. Li, and H. Liu, "Towards wearable a-mode ultrasound sensing for real-time finger motion recognition," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 26, no. 6, pp. 1199–1208, 2018.

[20] N. E. Bunderson and T. A. Kuiken, "Quantification of feature space changes with experience during electromyogram pattern recognition control," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 20, no. 3, pp. 239–246, 2012.

[21] J. Zeng, Y. Zhou, Y. Yang, J. Yan, and H. Liu, "Fatigue-sensitivity comparison of semg and a-mode ultrasound based hand gesture recognition," *IEEE Journal of Biomedical and Health Informatics*, vol. 26, no. 4, pp. 1718–1725, 2022.

[22] Z. Lu, S. Cai, B. Chen, Z. Liu, L. Guo, and L. Yao, "Wearable real-time gesture recognition scheme based on a-mode ultrasound," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 30, pp. 2623–2629, 2022.

[23] W. Xia, Y. Zhou, X. Yang, K. He, and H. Liu, "Toward portable hybrid surface electromyography/a-mode ultrasound sensing for human–machine interface," *IEEE Sensors Journal*, vol. 19, no. 13, pp. 5219–5228, 2019.